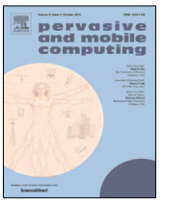
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Dynamic Partitioning and Popularity based Caching for Optimized Performance in content-centric fog networks: DPPCOP



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ABSTRACT

With the massive growth of information generation, processing, and distribution in the Internet of Things (IoT), the existing cloud architectures need to be designed more effectively using fog networks. The current IP-address-based Internet architecture is unable to deliver the desired Quality-of-Service (QoS) towards the increasing demands of fog networking-based applications. To this end, Content-Centric Networking (CCN) has been developed as the potential future Internet architecture. CCN provides name-based content delivery and is established as an architecture for next-generation fog applications. The CCN-based fog environment uses the cache of in-network fog nodes to place the contents near the end-user devices. Generally, the caching capacity of the fog nodes is very small as compared to the content catalog size. Therefore, efficient content placement decisions are vital for improving the network performance. To enhance the content retrieval performance for the end-users, a novel content caching scheme named "Dynamic Partitioning and Popularity based Caching for Optimized Performance (DPPCOP)" has been proposed in this paper. First, the proposed scheme partitions the fog network by grouping the fog nodes into non-overlapping partitions to improve content distributions in the network and to ensure efficient content placement decisions. During partitioning, the scheme uses the Elbow method to obtain the "good" number of partitions. Then, the DPPCOP scheme analyzes the partition's information along with the content popularity and distance metrics to place the popular contents near the end-user devices. Extensive simulations on realistic network topologies demonstrate the superiority of the DPPCOP caching strategy on existing schemes over various performance measurement parameters such as cache hit ratio, delay, and average network traffic load. This makes the proposed scheme suitable for next-generation CCN-based fog networks and the futuristic Internet architectures for industry 4.0.

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1. Introduction

The IoT has emerged as a rapidly growing architecture where IoT devices (end-users) are generating and consuming the contents [1]. Initially, the cloud computing systems have been conceptualized and developed to provide storage and processing capabilities at the centralized servers that interact with connected IoT devices [2]. However, presently,

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it becomes challenging to manage the increasing traffic needed for content distribution, using the existing cloud-based architectures due to the steep rise in the number of connected IoT devices and their applications. Cisco has forecasted that the number of IoT devices and their connections will grow to 12.3 billion by the year 2022 [3]. Moreover, several modern IoT applications such as gaming, health emergency, and public safety require real-time content retrieval with low latency from the cloud servers [4]. Due to immense growth in the number of IoT devices and the amount of content being required by them, the transmission of every content from the cloud servers leads to increased network traffic load and results in higher delay and degraded QoS [5–8].

Fog computing-based networking [9,10] has been developed as a promising computing architecture to fulfill the real-time demands of delay-sensitive IoT applications. In the fog environment, the network devices are called fog nodes that have storage and processing capabilities. These devices work as a transitional layer between the end-user devices and cloud servers. Thus, intermediate fog nodes can store the incoming contents from the cloud servers and provide services to its connected end-users with lesser latency [11]. Presently, the end-users are more focused on the rapid delivery of requested contents instead of searching the IP address of the cloud server/fog node to access the required content [12,13]. Towards this, the Content-Centric Networking (CCN) appeared to be a suitable architecture for the fog network environment. The CCN architecture decouples the content name from host locations and the end-users can access the required content using the content name in the fog network. The CCN-based fog environment supports in-network caching capability that distributes the content copies in the fog nodes to increase content availability across the network [14]. Caching of contents in the fog nodes reduces the network traffic load and results in efficient content delivery to the end-users [15]. Moreover, using content caching, the CCN would effectively manage the requirements of IoT applications and provide low-latency services to end-users [16].

As the caching capacity of fog nodes is very small as compared to the number of distinct contents in the fog network environment, efficient in-network caching is crucial for achieving optimal network performance. For this, the caching scheme essentially takes content placement (selection of in-network fog node for caching of content copy) and the replacement (removal of appropriate older content when the cache becomes full) decisions [17]. When each fog node takes caching decisions independent from the others, the same content can be excessively cached in many fog nodes that exist in the content delivery path and thus, increases the content redundancy. With excessive content redundancy in the fog nodes, when a cache miss occurs near the edges of the network, then there is a high probability that the upstream fog nodes also do not have the requested content as they have identical copies of contents in their storage as the downstream nodes. Therefore, the probability of retrieving the content from the cloud server increases which leads to performance degradation. Thus, there is a need for collaboration among nearby fog nodes to control excessive content redundancy and investigation of caching strategies to develop a CCN-based fog network that is scalable to the growing needs of the fog network environment.

The focus of this work is related to make effective utilization of the caching capacities of intermediate fog nodes (fog layer) between the end-user devices and the cloud servers. With efficient content placement decisions, the probability of accessing the content from the fog layer would increase. This would result in reduced content retrieval delay and QoS improvement for the end-user devices. In this direction, a novel caching scheme called Dynamic Partitioning and Popularity based Caching for Optimized Performance (DPPCOP) is proposed that is broadly divided into two parts. First, the scheme dynamically divides the fog network into the optimal number of partitions where each partition consists of several fog nodes and the end user devices. The partitions are formed for collaborative caching decisions in the fog layer to improve the content diversity in the network. During the cooperation, if the content requester and the provider exist within the same partition, then the content caching operation is not performed in any of the intermediate fog nodes inside the partition. This heuristic reduces the computational latency and content redundancy in the partition. When the content requester and provider belong to different partitions, then at most one content copy would be cached in each intermediate partition based on the proposed caching strategy. Secondly, for content caching operations, the proposed scheme selects the suitable fog node in each partition by considering content popularity and the fog node's distance-based metrics. The caching decisions based on the partition information, content popularity, and distance parameters place the popular contents near the end-users with high content diversity in the network. It has also been explored that the proposed scheme increases the network performance on quantitative parameters such as hit-ratio, average hop-count, and network traffic load *w.r.t* as compared to existing peer caching strategies. The summary of the contributions made in this paper are as follows:

- An adaptive partitioning scheme has been proposed that dynamically partitions the fog network to improve intra-partition content diversity. The scheme reduces the content replacement operations by caching the incoming content in at most one intra-partition fog node.
- With network partitioning, the content popularity, and distance from the content provider are jointly considered for the caching decisions. The caching probability escalates with an increase in content popularity and towards the edges of the fog network.
- The performance of DPPCOP has been evaluated in a discrete-event simulator on the realistic network topology and the results have been obtained for various QoS parameters. Extensive simulations illustrate the superiority of the DPPCOP scheme over the competing peer caching schemes in terms of hit ratio, average network delay, hop count, and traffic load.

The rest of the paper is organized as follows. A literature review related to the fog network environment and caching schemes has been discussed in the next section. Section 3 discuss the system model and assumptions. The proposed DPPCOP scheme is explained in Section 4 and the performance evaluation is performed in Section 5. The conclusion of the study is discussed in Section 6.

2. Literature review

Earlier, the cloud-computing-based architecture has evolved to provide computation and communication services to end-user devices using cloud data servers [18,19]. These cloud data servers are dispersed across geographical locations. Therefore, the computation–communication delay is highly dependent on the efficiency of the cloud servers and their distance from the end-users [20,21]. With the surge of IoT devices and their increasing content access requirements, the cloud servers faced severe performance issues for real-time IoT applications [22].

To mitigate the limitations of cloud architecture and deliver real-time services to modern IoT applications, Cisco has proposed the fog-computing architecture [10]. In the fog architecture, the fog nodes deliver the requested services, which are comprised of network routers and gateways. The detailed significance of the fog-network environment for IoT has been discussed in [23,24] in terms of delay, mobility, and geographical dispersion. As stated in [25], the modern IoT applications require moderate to high storage space within the fog networks and therefore these applications need a competent content management mechanism. The CCN architecture can effectively utilize the available resources in fog-networking and can be scaled to meet the increasing demand for IoT devices [26]. In CCN-based fog networks, the selection of suitable fog nodes for content caching operations plays an important role in reducing the average network traffic, and load from the server and improving QoS for end-users.

Initially, content caching has attracted the attention of several researchers in the Radio Access Networks (RAN) and Metro Access Networks (MAN) before the deployment of the caching schemes in fog networks. Piao et al. [27] provides a comprehensive survey of the caching schemes for RAN and summarized various edge cache content placement strategies. The results of the work show that efficient caching decisions are crucial for improved network performance. It also analyzes the significance of content popularity, content size, content type, and several other parameters during caching decisions. A game-theoretic approach for caching has been presented in [28] for the Fog-Radio Access Networks (F-RANs). The scheme improves the cache space utilization in the fog access points using a hierarchical resource management architecture. The scheme organizes the fog nodes into multiple clusters for cooperation and proposed two reinforcement learning algorithms to maximize the long-term utility by analyzing the channel gains and content requests. Savi et al. [29] discussed the need for cooperation between the fixed MANs and the mobile networks. The work suggested an Integer Linear Programming (ILP) model for the fixed network topology to optimize content placement and Video-on-Demand (VoD) content routing. Ayoub et al. [30] further suggested an ILP model to improve caching performance for Video-on-Demand (VoD) contents in the metro area networks. The scheme minimizes the energy consumption using a heuristic algorithm that power-on and -off the cache dynamically and uses the suggested caching strategy. The scheme addressed the issue of the Trombone effect and deduce the need for a new architecture for the networks.

Recently, various in-network caching schemes are presented specifically for the CCN also. These caching schemes are divided into the off-path and on-path caching strategies [31]. The on-path caching schemes place the content copy in those fog nodes that appear in the content delivery path towards the requester. On the other hand, the off-path caching schemes can place the contents in those fog nodes also that are not forwarding the content towards the requester. While the on-path caching is an integral part of the CCN, the off-path caching requires centralized network management. Hence, on-path caching strategies are widely implemented due to lesser communication overhead, processing latency and network traffic in contrast to off-path caching schemes [32].

The conventional content placement scheme in the CCN is Leave-Copy Everywhere with Least-Recently-Used (LRU) replacement strategy [33]. In this scheme, each on-path router cache a replica of the content. However, the scheme does not utilize the available caching capacity comprehensively and excessive caching operations cause higher operational costs [34]. Therefore, a better selection of the network devices for content caching operations would improve the network performance. For this, various studies have been proposed recently that are taking different parameters for content caching decisions. The Leave-Copy Down (LCD) caching approach [35] is intended to decrease redundancy in the network where the content provider places the copy of the content on the downstream router only. A variation of the LCD scheme, called Move-Copy Down (MCD) moves the content towards end-users with the increase in content popularity and further decreases content redundancy as compared to the LCD strategy. A distance-based caching approach is suggested in [36], which effectively multiplex the contents of different traffic flows. The scheme increases the caching probability with an increase in the distance traversed by the content.

The "Cache less for more" strategy [37] considers node betweenness centrality parameter for content caching operations and shows significant performance improvement from random probabilistic (RandProb) caching scheme [38]. Several centrality-based caching schemes are discussed in [39,40] that analyze the effects of node degree, betweenness, stress, closeness, and graph centrality-based caching decisions on network performance. The work shows that node degree centrality (DC) is a simple and effective criterion for content placement decisions. The authors also concluded that the heterogeneous caching capacity of network routers has limited gain in QoS parameters for the end-users. The multi-attribute caching approaches, named CPNDD (Content Placement based on Normalized Node Degree and Distance) are

proposed in [41,42] that analyze the joint effect of node degree and hop-count parameters on caching performance and show a significant gain in the hit-ratio from the LCE, RandProb, and Degree Centrality (DC) based caching schemes.

Parallel to the above-mentioned schemes, several strategies argue that caching of popular contents also achieves high network performance and reduces content replacement operations [43,44]. In the MPC (Most Popular Content) caching scheme [45], the network routers identify popular contents based on their access patterns and suggest these contents to the nearby routers for caching. The MAGIC (MAx-Gain In-network Caching) [46] caching strategy performs caching operations in the on-path routers after considering content popularity, hop-reduction, and caching penalty parameters. However, the scheme suffers from high computational delay and hence, is not suitable for real-time fog applications. The FGPC (Fine-Grained Popularity-Based Caching) scheme [47] cache the contents based on their popularity and LRU replacement strategy. Yet, due to the static threshold value for the recognition of popular content, the scheme fails to comprehensively utilize the available caching resources. The CPUL (Content Popularity and User Location) [43] classify the contents into popular and unpopular and suggested different caching schemes for them. However, the scheme implemented a centralized popularity determination mechanism, which leads to higher communication and computational delay in large-scale networks. The scheme mentioned in [32] uses a large popularity window for content popularity analysis and takes content placement decisions based on the content popularity, content access pattern, and caching capacity of the network routers. A multi-attribute-based caching scheme called *Opt-Cache* is suggested in [48] that jointly considers the content popularity and hop reduction parameters during the caching decisions. The scheme places the popular contents in the network nodes for longer periods and reduces the bandwidth consumption and the frequency of caching operations.

To reduce the average network traffic and server load, various network partitioning-based caching schemes have also been proposed for the CCN. Each network partition contains several nodes and reduces the excessive content redundancy using cooperative caching decisions within the partitions during content message forwarding [49]. The HCC (Hierarchical Cluster-based Caching) strategy [50] divided the network into two layers hierarchically. The partitions in the core layer do not have the content caching capability and edge partitions cache the contents based on betweenness centrality parameter. The scheme mentioned in [51] partitioned the network into a fixed number of clusters and considers content popularity for caching decisions. It selects the partition heads for each cluster based on their storage capacities as compared with the other nodes in the partition. However, due to random partition formation and naive head selection procedures, the scheme fails to achieve optimal network performance. A hierarchical partitioning and distance-based caching strategy is presented in [52]. The scheme divided the network routers into the backbone layer and the intermediate layer. Both layers are capable to cache the incoming contents and takes caching decisions based on the distance traversed by the requests and their matching content messages. However, the scheme does not consider the content retrieval pattern during content placement, which results in poor utilization of caching resources as rarely accessed contents are also cached by the on-path routers. The scheme mentioned in [53] designed a caching strategy based on the Cluster-based Scalable Router (CSR). Each CSR consists of several routers that are externally seen as a single router and internally deployed a load-balancing strategy among them. Each cluster uses a load balancer component to cache the content on the routers. However, due to centralized caching and distribution operations, the scheme suffers from significant computational and communication overheads.

A popularity-driven coordinated caching scheme is proposed in [54]. The scheme determines the content popularity using the historical request statistics in a fixed period. It formulates the problem-solving models to minimize average network traffic load and hop count metrics. The authors presented two popularity-based caching strategies called *TopDown* and *SympOpt* that take coordinated caching decisions in the content delivery path. However, due to a lack of coordination among nearby nodes that do not exist in the content delivery path, the cache space of in-network routers is not utilized comprehensively and has further scope for improvement.

An intra-AS cooperative caching strategy is suggested in [55] that eliminates excessive content redundancy in the network and collaboratively serves the requests. The scheme periodically removes the redundant contents from the network cache and thus increases the communication overhead significantly. Moreover, the scope of collaboration among network nodes is limited to one-hop neighbors only which may create a large number of autonomous systems in the network and would influence the scalability of the system.

Another variation of the cooperative caching scheme [56] is specifically suggested for the video stream delivery in which contents have many segments that are accessed consecutively. The scheme jointly considers the traditional hash-based and directory-based coordinated caching schemes during content placement decisions. During simulation executions, the scheme shows coordination among a random number of network routers for the caching decisions. Moreover, the scheme may not improve the caching performance in those network systems where contents are not accessed sequentially and are not dependent on each other.

When available caching capacity becomes full, the network node uses the content replacement algorithms during caching operations [57]. Generally, the LRU, First In First Out (FIFO) and Least Frequently Used (LFU) replacement algorithms are implemented with content placement strategies to evict the older contents from the cache to make space for incoming ones. From these replacement algorithms, LRU becomes a popular choice because it replaces the contents which have not been used for the longest period, and therefore it is sensitive to dynamic popularity trends of contents as compared to peer techniques [58].

From the existing literature work, it is evident that the efficient content caching mechanism is vital for improving the QoS in a CCN-based fog network environment. Moreover, the selection of a suitable fog node for content caching is crucial

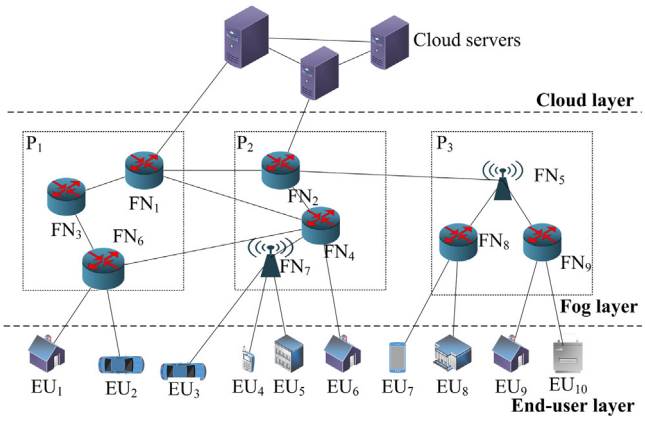


Fig. 1. An illustration of network model.

to optimize network traffic load and latency in retrieving the requested content because that fog node would serve the future content requests for a longer duration. To the best of our investigation in the literature, no scheme proposed a partitioning-based caching scheme for content-centric fog networks that considers content and network characteristics during content placement decisions. With this motivation, a novel content caching scheme named DPPCOP is proposed in this paper that partitioned the fog network into a sufficiently "good" number of partitions and takes content placement decisions using the content popularity and the distance metrics. In DPPCOP, the intra-partition fog nodes collaborate to increase the in-network cache hit ratio. To reduce the excessive caching operations, at most one intra-partition fog node would cache the contents that are arriving from other partitions. The detailed system model of the DPPCOP scheme is defined in the following section.

3. System model

3.1. Overview of CCN-based fog network

The CCN uses a name-based content access mechanism and the content is retrieved from the cache of the nearest available fog node. For CCN operations, each fog node maintains three tables called Content Store (CS), Pending Interest Table (PIT) and Forward Information Base (FIB) [12]. The CS caches the incoming *Content* messages based on the caching scheme. The PIT stores the information of *Interest* messages that are forwarded to upstream fog nodes for content retrieval. The FIB maintains information of the fog nodes to forward the *Interest* message. The end-user device generates the *Interest* message that has the name of the required content. This message is forwarded hop by hop until it reaches the cloud server or the required *Content* found in the cache of the intermediate fog node. On arrival of an *Interest* message, the fog node first explores its CS for the required content. If the *Content* exists in the CS, then the fog node forwards the corresponding *Content* message towards the end-user on the reverse path. If the content is not available in the cache, then the fog node creates an entry for *Interest* message in the PIT and forwards it towards the suitable upstream fog node based on the information from FIB. The *Interest* message forwarding process ends when the requested content found in the CS of any on-path fog node or the *Interest* message reaches the content server (Cloud server) that has all the contents in its storage. When the on-path fog node receives a *Content* message, it checks the PIT to forward the *Content* message to matching end-users and then takes the caching decisions. If no matching PIT entry exists, then the fog node simply discards the message without further forwarding in the network.

3.2. Network model

Fig. 1 illustrates a model of CCN-based fog network as a graph $G(V, E)$. In the model, V represents the set of network nodes that includes the cloud servers $(serv_1, \ldots, serv_s)$, fog nodes $(FN_1, \ldots, FN_n)$, and the end-user IoT devices $(EU_1, \ldots, EU_m)$. The E represents the collection of links among network nodes. The model divides the fog network into three layers named cloud layer, fog layer, and end-user layer. The cloud layer is connected with the end-user layer using multi-hop links passing through the fog layer. Initially, all the distinct contents $(D_1, D_2, \ldots, D_{|D|})$ are stored in the cloud servers only. The end-users do not have the caching capability. But, the in-network fog nodes can cache the contents based on the underlying caching policy, which enables them to satisfy the subsequent *Interest* messages from their cache storage. Without losing generality, it has been assumed that the fog nodes e.g. network routers and access points in the fog layer have identical caching capacity, and the payload size in *Content* messages is also uniform. To model the real-world network environment, the caching capacity of fog nodes is kept very small as compared to content catalog size, which can be shown as follows:

$$|CS(FN_i)| \ll |D| \tag{1}$$

Table 1
Variables definition.

Variable	Definition
$CS(FN_i)$	Content store of fog node FN_i.
$\|CS(FN_i)\|$	Content store size of fog node FN_i.
$Name(I_i)$	Name of requested content in *Interest* message I_i.
$\|D\|$	Number of distinct contents in the network (content catalog size).
$Name(D_i)$	Name of content in *Content* message D_i corresponding to *Interest* message I_i.
$W(FN_i)$	Current popularity window for fog-node FN_i.
$W_{D_m}(FN_i)$	Number of occurrences of *Content* message D_m in the popularity window of FN_i ($W(FN_i)$).
$Size(W(FN_i))$	Current popularity window size for fog-node FN_i.
$Max(Size(W(FN_i)))$	Maximum permissible popularity window size for fog node FN_i.
λ	Interest rate.
T_{FN_i}	Threshold value for content placement decision in FN_i.
α	Exponent parameter in Zipf distribution.
$PH(FN_i)$	Partition head of fog node FN_i.
$PH(P_m)$	Partition head for the partition P_m in the fog network.
$Dist(I_k)$	Distance traversed by the *Interest* message I_k.
$Dist(D_k)$	Distance traversed by the *Content* message D_k.
B	Average connections bandwidth among fog nodes in the Interest forwarding path.
$H(P_x, P_y)$	Number of hops in the shortest path between the farthest fog-nodes belonging to partitions P_x and P_y.
$H(FN_j, PH(FN_j))$	Number of hops in the shortest path between the fog node FN_j and partition head of FN_j.
$Dist(FN_i, FN_j/Serv)$	Number of hops in the shortest path between the fog node FN_i and $FN_j/Serv$.
$Dist(P_x, P_y)$	Shortest path distance between the farthest fog nodes $FN_i \in P_x$ and $FN_j \in P_y$.
$Dist(FN_j, PH(FN_j))$	Shortest path distance between the fog node FN_j and partition head of FN_j.
$Dist(FN_i, FN_j/Serv)$	Shortest path distance between the fog node FN_i and $FN_j/Serv$.
$b_width(P_x, P_y)$	Minimum bandwidth in the shortest path between farthest fog-nodes of partitions P_x and P_y.
$S.PH(D_i)$	Partition head name of source/content provider.
$O.PH(D_i)$	Partition head name of the last-encountered on-path fog-node that forwarded the *Content* message (D_i).
$\phi(D_i)$	Control the caching decisions for D_i.

3.3. Dynamic model and assumptions

To control the number of content caching operations, the network $G(V, E)$ has been partitioned into the different non-overlapping groups (partitions) $(P_1, P_2, \ldots, P_p)$ using the proposed partitioning strategy. Instead of randomly fixing the number of partitions initially as suggested in the k-means clustering algorithm, the proposed network partitioning scheme uses a bottom-up approach to heuristically compute the "sufficiently good" number of partitions in the network. Therefore, the proposed network partitioning strategy creates a distinct number of partitions for different network topologies dynamically and can adapt to different network configurations. The detailed significance of the proposed partitioning approach has been discussed in the next section. For modeling of *Interest* message arrival patterns, the content popularity follows a Zipf distribution, which is a widely accepted model for current Internet applications [32,39]. The Zipf distribution shows non-uniform access patterns for different contents and the contents are ranked based on their popularity. Sometimes, the requesters generate *Interest* messages for the sporadic contents that are hardly ever accessed again and are cached for extended periods in the fog layer. It results in poor utilization of the available cache resources. Therefore, the proposed scheme considers the popularity of contents during caching decisions to reduce excessive content replacement operations for unpopular content. Based on the location of fog nodes in the network, each fog node may encounter a different number of *Interest* messages per unit time and hence, time duration cannot be used to determine content popularity reliably. Therefore, instead of computing the content popularity based on *Interest* messages received in previous "s" seconds, each fog node maintains a novel dynamic popularity window to determine the popularity of contents.

On arrival of an *Interest* message, the popularity window stores only the name of the requested content and thus has a limited overhead on the available cache space. It has been explored in the existing research works [36,42] that the network performance improves when the contents are cached near the edges of the network. In the proposed caching scheme, the popularity of a content has been determined from the proposed popularity window by counting the access requests for that specific content. This characteristic is considered along with the fog node's distance during the content placement decisions. By using these parameters, the caching probability of popular contents is improved with an increase in the distance of the fog node from the content provider and hence, the popular contents would be placed near end-users with high probability. Once the CS of a fog node becomes full, the proposed scheme uses the LRU replacement scheme for content evictions. To simplify the further discussion, the variables and definitions used in the paper are defined in Table 1.

3.4. Updated Interest and Content message structures

The proposed caching scheme uses partitioning information along with content popularity and distance metrics during caching operations. To store this information, the structure of the *Interest* and *Content* messages has been modified as follows.

The updated structure of the *Interest* message contains the $Dist(I_i)$ additional field in the traditional message [33]. The $Dist(I_i)$ field stores the distance traversed by the *Interest* message towards the content provider.

For efficient caching decisions, the *Content* message has been updated with the following additional fields; $Dist(I_i)$, $Dist(D_i)$, $S.PH(D_i)$, $O.PH(D_i)$ and $\phi(D_i)$ along with requested payload. The content provider replicate the value of $Dist(I_i)$ from *Interest* message (I_i) to *Content* message (D_i). The Section 4.1 proposes a novel network partitioning scheme that divides the network into a number of partitions and each network node is associated with a partition head. The partition heads are created to control the content caching decisions and are discussed in the subsequent section. During content message preparation, the content provider (fog node/server) mentions the name of its partition head in the $S.PH(D_i)$ field and each on-path fog node (FN_k) places its partition head name in the $O.PH(D_i)$ field before forwarding of *Content* message towards end-user devices. The value of the field, $\phi(D_i)$ is set or reset to enable/disable the content caching operations.

4. Proposed scheme: DPPCOP

This section elaborates the proposed network partitioning scheme and partition head selection mechanism along with novel *Interest* and *Content* message forwarding/caching mechanisms.

4.1. Proposed network partitioning scheme

In the proposed scheme, initially, each fog node has been designated as a partition head. Then, the proposed scheme iteratively decreases the number of partitions by merging of two closest partitions. In the proposed partitioning scheme, the closeness of the partitions has been determined based on the bandwidth and hop-count parameters. Moreover, the proposed partitioning scheme optimizes the number of partitions using the Elbow method [59] and performs the "Network_Partitioning (n, k, P_x, P_y)" procedure in the fog layer having n-fog nodes for network partitioning. Here "n" represents the total number of fog nodes in the network. The parameter "k" denotes the total number of non-overlapping partitions that are created in the network after the execution of the algorithm. The P_x and P_y illustrate the distinct and non-overlapping partitions in the fog network having one or more fog nodes.

Network_Partitioning (n, k, P_x, P_y)

Initialize each $\{FN_x; 1 \leq x \leq n\}$ as P_x with $PH(P_x) = FN_x$.
Update_Distance(P_x, P_y)

1. Do $x := 1$ to n
2. Do $y := 1$ to x
   (a) Compute, $Dist(P_x, P_y)$ using Equation (2)
   (b) $M(P_x, P_y) := Dist(P_x, P_y)$
3. End do
4. End do

Merge_Partition (P_x, P_y)

1. Determine WSS_k using Equation (3) for initial n-partitions
2. i:=0
3. Do
   (a) i:=i+1
   (b) Select P_x and P_y, such that $Dist(P_x, P_y)$ is minimum in $M(P_x, P_y)$
   (c) Merge P_x and P_y to create $P_{x,y}$ to form $(k-1)$ partitions in the network
   (d) Determine WSS_{k-1} using Equation (3)
   (e) If $\left(WSS_{(k-1)} - WSS_k > \delta\right)$, then
      i. Discard merging of partitions performed in $step-3(c)$
      ii. Stop iteration of $step-3$
      iii. End If
      iv. End do
   (f) Else,
      i. k:= k-1

ii. $PH(P_{x,y}) := FN_i$, where $FN_i \in P_{x,y}$ and $Dist(FN_i, Serv)$ is minimum
iii. Execute *Update_Distance*$(P_{x,y}, P_z)$ procedure, for any other partition P_z
iv. End

In the proposed "Network_Partitioning (n, k, P_x, P_y)", initially each fog node (FN_x) has been designated as a partition P_x and its own partition head $(PH(P_x))$. In the first iteration, the proposed scheme uses the "*Update_Distance*(P_x, P_y)" procedure to compute the distance among fog nodes. In *step* − 1 *and step* − 2 of "*Update_Distance*(P_x, P_y)" procedure, the scheme considers the average bandwidth among disjoint partitions, bandwidth between P_x and P_y and hop-count for distance determination and placed the values in the matrix $M(P_x, P_y)$. This matrix $M(P_x, P_y)$ is computed each time the merging of partitions is performed and it is used to determine the two closest partitions for the merge operation. The expression to determine $Dist(P_x, P_y)$ is shown in Eq. (2).

$$Dist(P_x, P_y) = \frac{B_width}{b_width(P_x, P_y)} \times H(P_x, P_y) \tag{2}$$

The distance between two partitions has been computed as the ratio of the product of average network bandwidth (B_width) with the shortest path (hop-count) between two farthest fog nodes $(H(P_x, P_y))$ in the partitions and the minimum bandwidth between those fog nodes $(b_width(P_x, P_y))$. By using this approach, tightly coupled partitions are formed. The proposed scheme iteratively selects two closest partitions from $M(P_x, P_y)$ and perform their merging using *Merge_Partition*(P_x, P_y) procedure. As mentioned in *step*−1 of the *Merge_Partition* procedure, the value of WSS_k is computed using Eq. (3).

$$WSS_k = \sum_{j=1}^{n} Dist(FN_j, PH(FN_j))^2 = \sum_{j=1}^{n} \left(\frac{B_width}{b_width(FN_j, PH(FN_j))} \times H(FN_j, PH(FN_j)) \right)^2 \tag{3}$$

Here, parameters *"k"* and *"n"* depict the number of partitions and the total number of fog nodes in the network respectively. The equation computes the distance of the fog nodes from their partition heads respectively. It uses the following parameters during computation of WSS_k; average network bandwidth, minimum connection bandwidth between fog node and its partition head $(b_width(FN_j, PH(FN_j)))$ and the number of hops in the shortest path between the fog node and its partition head $(H(FN_j, PH(FN_j)))$. Initially, when each fog node itself is a partition head $(k = n)$, then the value of parameter $Dist(FN_j, PH(FN_j))^2$ would become 0 for all fog nodes. Hence, the value of WSS_k would be *0*. Consequently, if the network nodes are connected using identical bandwidth connections then after merging of two partitions, the remaining partitions would be *(k = (n−1))* and the value of WSS_k would be computed as 1 and so on.

In *step* − 3(*a*) − 3(*c*), the partitions P_x and P_y are selected from $M(P_x, P_y)$ that are closest to each other and are merged to form $P_{x,y}$. Each iteration of this step would decrease the total number of partitions in the network by 1. To find the optimal number of partitions in the proposed partitioning scheme, the Elbow method [59] has been used. It determines the change in the value of the Within-cluster Sum of Squares (WSS) by decreasing the total number of partitions and a curve has been plotted for these values according to the number of partitions. Eq. (3) shows the expression for WSS computation [60].

Initially, the value of the WSS parameter does not increase sharply with the merging of partitions but the value shows a rapid increase after a certain number of partitions are created and then forms an "Elbow". This bend is used as an indicator of the optimal number of partitions. For this, the scheme computes WSS_{k-1} after each merge operation and determines the gain achieved from the WSS_k as mentioned in *step* − 3(*e*). If the gain is greater than δ, it means that the "Elbow" has been formed, and therefore, the newly formed partition would be discarded, and the merging of partitions stops with *"k"*- partitions. Otherwise, as shown in *step* − 3(*f*), the intra-partition fog node, closest to the cloud server, would be designated as the new partition head, which plays a crucial role in controlling content caching operations. Its significance has been elaborated in Section 4.3 in detail. Finally, the distance between the newly formed partition and the remaining partitions is computed and a new matrix $M(P_x, P_y)$ is formed (*step* − 3(*f* − *ii*)).

For illustration purpose, Fig. 1 shows three partitions P_1, P_2 and P_3. In these partitions, the fog nodes FN_1, FN_2, and FN_5 are designated as partition heads using the proposed partitioning scheme as they are nearest to the cloud servers. The partition heads assist in controlling the content redundancy and replacement operations within partitions.

4.2. Interest message handling procedure

When a fog node FN_j receives an *Interest* message (I_i) which generated by end-user IoT device (EU_k), it uses *"Interest message handling procedure"* for *Interest* message processing. In *step-1* and *step-2* of the procedure, the FN_j determine the size of its popularity window and places the name of requested content $(Name(I_i))$ into it. If a free slot is available in the

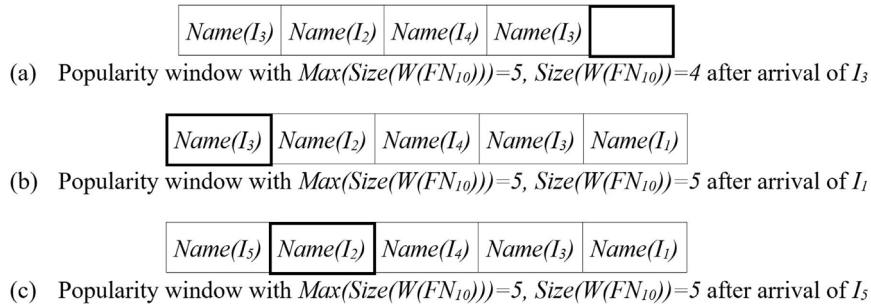


Fig. 2. An illustration of the popularity window with $Max(Size(W(FN_j))) = 5$.

popularity window, then the ($Name(I_i)$) is placed into that slot. Otherwise, the FIFO replacement scheme is used to evict the older requests when the popularity window becomes full.

An illustration of the popularity window for a fog node (FN_{10}) is shown in Fig. 2 to explain its working. In this example, the next slot for storing the name of the requested content is highlighted using the thick border. Fig. 2(a) shows the state of popularity window after arrival of *Interest* message I_3. As two empty slots were available, the $Name(I_3)$ is stored in the 4th slot of the popularity window. Consequently, when *Interest* message I_1 arrives, the FN_{10} determines that one more slot is empty (highlighted using thick border) as shown in Fig. 2(a) and therefore, its name is stored in the empty slot (Fig. 2(b)). When the popularity window reaches its maximum capacity, the pointer to next slot for FIFO replacement is updated and highlighted in Fig. 2(b) using thick border. Hence, during processing of subsequent incoming *Interest* message I_5, the $Name(I_3)$ is replaced by $Name(I_5)$ and the pointer to next slot for the FIFO replacement is updated as shown in Fig. 2(c).

Procedure-1: Interest message handling procedure (I_i, FN_j, EU_k)

Perform following operations on arrival of *Interest* message (I_i) from EU_k at node FN_j

1. If $Max(Size(W(FN_j))) > Size(W(FN_j))$ then Replicate $Name(I_i)$ to empty slot of $W(FN_j)$
2. Else, Replicate $Name(I_i)$ from I_i to $W(FN_j)$ using FIFO replacement
3. Update $Dist(I_i)$ field in I_i as:
   (a) $Dist(I_i) := \frac{B}{b_width(FN_j, EU_k)} \times H(FN_j, EU_k)$
4. If requested Content exists in the $CS(FN_j)$, then go to *Content message handling procedure*
5. Else, if I_i exists in PIT(FN_j), then aggregate I_i in PIT(FN_j) and dispose I_i
6. Else, if FIB(FN_j) has upstream fog-node/server information for I_i forwarding, then forward I_i to upstream fog-node/server accordingly
   (a) Make record of I_i in $PIT(FN_j)$
7. Else, dispose I_i from the fog-network
8. End

The fog node then updates the $Dist(I_i)$ field in the *Interest* message as shown in *step-3* which uses average network bandwidth in the fog layer, hop-count, and minimum bandwidth between FN_j and EU_k. If the required content found in the cache of FN_i, then the *Content message handling procedure* would be executed as shown in *step-4*. Otherwise, *step-5* to *step-7* are executed by FN_i to perform traditional *Interest* message processing steps as discussed in Section 3.1.

4.3. Content message handling procedure

When the requested content name, $Name(I_i)$ is found in the cache of fog node (FN_j) or *Interest* message reaches to the cloud server, the node becomes content provider and it prepares a *Content* message D_i as shown in *"Content message handling procedure"*. The content provider (FN_j) uses *step-1* for the message preparation process and copy the value of $Dist(I_i)$ field from I_i into the identical field of *Content* message D_i. Then, the (FN_j) initializes its partition head name in the $S.PH(D_i)$ and $O.PH(D_i)$ fields of D_i. Then, the value of $\phi(D_i)$ has been reset to 0 and the *Content* message is forwarded in the reverse direction towards the requester (EU_k). When on-path fog node (FN_m) receives the D_i, it update the $Dist(D_i)$ field to the distance between FN_j and FN_m as shown in *step-3*. If the on-path fog node (FN_m) is a partition head, then

it updates its name in the $O.PH(D_i)$ field. If (FN_m) is outside the partition of content provider *(step-4(b))*, then it set the $\phi(D_i) = 1$ to enable content caching within its partition. For efficient content placement decisions, FN_m determines the popularity of incoming content by counting its occurrences in the popularity window and computes ratio of the distance traversed by D_i and I_i. Using these parameters, the scheme computes the *Placement_Score* as mentioned in *step-5*. The value of *Placement_Score* increases with an improvement in the content popularity and increase in the distance traversed by the content message. Thus, the proposed scheme places the popular contents near end-user devices. *Step-6* ensures that at most one copy of D_i would be placed in each partition in the content delivery path where *Placement_Score* is greater than or equal to the threshold value. According to *Step-6(b)*, once the content is cached within a partition, the value of $\phi(D_i)$ is reset to 0, which can again be set to 1 by the next partition head only in the delivery path. Therefore, rest of the fog nodes within that partition (where caching is performed) cannot cache the incoming content because $\phi(D_i) = 0$. The caching of incoming content would be disabled until $\phi(D_i)$ is set to 1 as mentioned in *step-6* which is possible only when next partition head would receive the incoming *Content message*. If the cache of FN_m was full, then the oldest content is evicted using the LRU replacement policy and finally, the *Content* message is forwarded towards requesters based on the information mentioned in the PIT of FN_m as shown in *step-7*.

Procedure-2: Content message handling procedure (D_i, FN_j, FN_m, EU_k)

1. If requested content exists in the $CS(FN_j)$ then
   (a) Create *Content* message (D_i) corresponding to I_i
   (b) Copy the value of $Dist(I_i)$ field from *Interest* message I_i to D_i
   (c) Set the value of field, $Dist(D_i)$:=0
   (d) Set the name of $PH(FN_j)$ in the $S.PH(D_i)$ and $O.PH(D_i)$ fields of *Content* message
   (e) Reset the field, $\phi(D_i) := 0$
   (f) Forward D_i in the reverse direction towards EU_k.
   (g) End If
2. Else, when *Content* message is received at the intermediate fog node FN_m then perform *step-3 to step-7*.
3. Overwrite the value in distance field $Dist(D_i)$ of D_i as distance between FN_j and FN_m
   $Dist(D_i) := \frac{B}{b_width(FN_j, FN_m)} \times H(FN_j, FN_m)$
4. If, FN_m is a partition head Then,
   (a) Overwrite its name (FN_m) into $O.PH(D_i)$ field.
   (b) If $S.PH(D_i)! = O.PH(D_i)$ then
   Set the field, $\phi(D_i) := 1$.
   End If
5. Compute, $Placement_Score := W_{D_i}(FN_m) \times \frac{Dist(D_i)}{Dist(I_i)}$
6. If $\phi(D_i)$=1 & $O.PH(D_i) = PH(FN_m)$ & $S.PH(D_i)! = PH(FN_m)$ & $Placement_Score \geq T_{FN}$, then
   (a) Cache content in the $CS(FN_m)$ using the LRU replacement mechanism
   (b) $\phi(D_i)$:=0
   (c) End If
7. Forward content towards IoT device EU_k

5. Performance evaluation

This section first elaborates the simulation environment parameters and then presents the simulation results. During simulation executions, the proposed DPPCOP strategy is compared with several state-of-the-art caching strategies, named LCE [33], LCD [35], Degree-Centrality (DC) based [39], FGPC [47], and CPNDD [42] approaches. Additionally, the proposed caching scheme is also compared to the scenario where fog nodes have no caching capability (NoCache scheme) on performance parameters such as average network hop count, delay, and network traffic load.

5.1. Simulation parameters and setup

To obtain realistic simulation results, the performance of caching strategies is examined on the Abilene-based topology [61] with 167 nodes and another topology similar to the US26 network [62] with 26 fog nodes. The illustration of

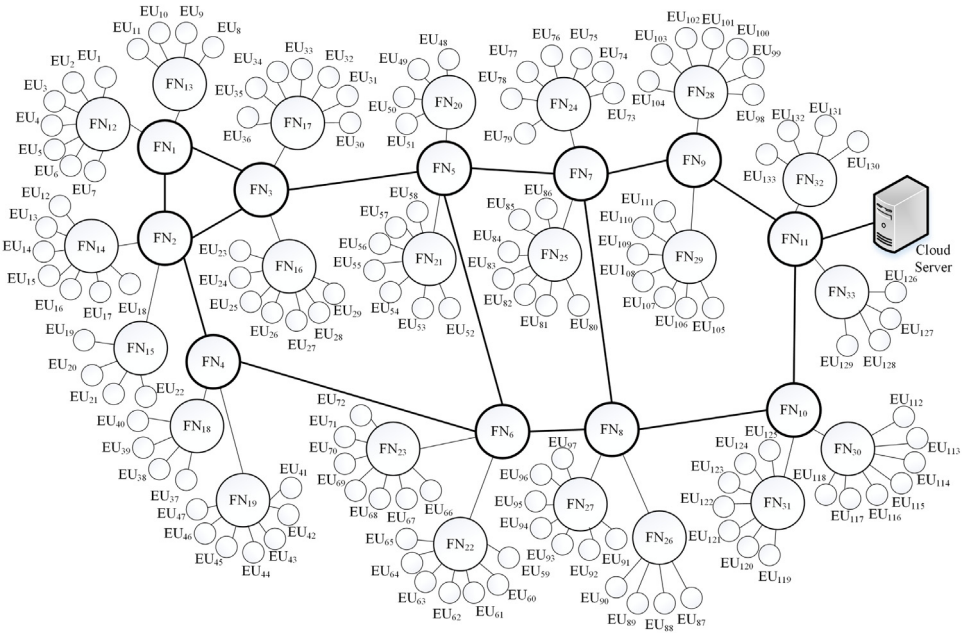


Fig. 3. An illustration of Abilene-based network topology.

the Abilene-based network topology is shown in Fig. 3 that consists of 33 fog nodes (FN_1, FN_2, ..., FN_{33}), 133 end-user IoT devices (EU_1, EU_2 ..., EU_{133}) and a cloud server. As shown in Fig. 3, the Abilene-based network consists of 11 backbone fog nodes {FN_1, FN_2, ..., FN_{11}}.

The simulation executions have been performed in the ndnSIM simulation environment [63]. In the beginning, the cloud servers hold the entire content catalog in both network topologies. The caching schemes are implemented in the fog layer of the architecture shown in Fig. 1. The end-users do not have caching capability and they are the source of *Interest* messages. For Interest message routing, the best-route routing mechanism has been implemented during the evaluation of caching schemes. This routing strategy forwards the Interest messages to upstream fog nodes with the lowest routing cost [64] by analyzing the Green–Yellow–Red interfaces in the network [65]. An interface is categorized as "Green" if required content flows back from that interface. The interface turned "Yellow", if the pending interest is timed out. The interfaces that cannot bring the required content back to the requester are defined as red interfaces. In the best-route routing mechanism, "Green" interfaces have the highest ranking for interest message forwarding to minimize the routing cost.

The content catalog size ($|D|$) and caching capacity of fog nodes ($|CS(FN_i)|$) has been set to *5000* and *50* ($|CS(FN_i)| \ll |D|$). Thus, the caching capacity of the entire fog layer would be 1650 (33 (number of fog nodes) $\times$ 50 (caching capacity of each fog node)) and 1300 (26 $\times$ 50) in the Abilene and US26-based network topologies. Each *Content* message carries *1 KB* of payload in response to the *Interest* message.

The value of exponent attribute (α) (popularity skewness parameter) in the Zipf distribution, is set to *0.8 and 1.1* under different simulation executions to examine the performance of caching schemes under a wide variety of content popularity distributions. In the Zipf model, α defines the skewness in the Interest message arrival patterns. Eq. (4) shows the normalized access frequency for the xth rank content in Zipf distribution. According to the equation, with an increase in the value of α, the distribution becomes more skewed and the access probability for the 1st rank content becomes closer to 1. Thus, when α increases in Zipf distribution, a smaller fraction of content catalog ($|D|$) is accessed more frequently as compared to other contents [31]. Therefore, when $\alpha = 0.8$, then the content access pattern would be more uniform as compared to $\alpha = 1.1$ and larger cache capacities would be required in the fog layer to place popular contents.

$$f(x; \alpha, |D|) = \frac{1/x^{\alpha}}{\sum_{n=1}^{|D|}(1/n)^{\alpha}} \tag{4}$$

The request rate (λ) is *50/s* per End-user device and the LRU has been assumed as the default content replacement strategy in all schemes. To determine the popularity of a content, the popularity window of each fog node stores up to *500* name of contents that are requested by the end-user devices. Hence, the popularity of a content is determined by analyzing the previous 500 requests that are received at the fog node. Based on the content popularity, frequently accessed contents are placed in the cache of fog node ($CS(FN_i)$). The simulation results start showing stabilization within the first 50 s for both network topologies and therefore the simulations have been performed for *150 s*. This is due to

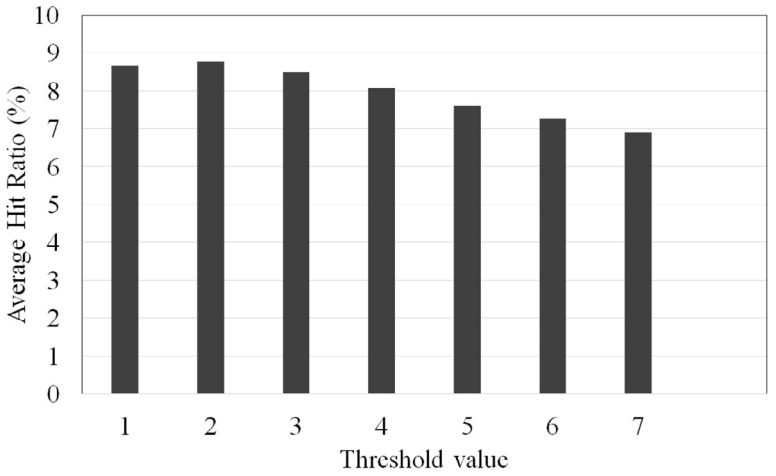


Fig. 4. Optimal threshold value (T_{FN}) determination for the DPPCOP scheme.

the fact that with 133 end-user devices in the Abilene-based network and $\lambda = 50/s$, approximately $(133 \times 50) = 6650$ *Interest* messages are generated every second. Similarly, nearly 1300 *Interest* messages are created every second in the US26-based network. To maintain uniformity in the network, the link delay and bandwidth among the network nodes is kept to *10 ms* and *10 Mbps* respectively.

Before performing the caching operations, the proposed network partitioning scheme first divides the fog layer of the networks into a sufficiently "good" number of partitions. During partitioning of Abilene and US26-based network topologies, the value of δ has been set to 30 which is used to represent "Elbow" during computation of WSS_k parameter. It indicates that the partitioning scheme would stop merging of the partitions when the difference between $WSS_{(k-1)}$ and WSS_k becomes greater than 30. Initially, the value of $WSS_{(k-1)}$ does not increase sharply with the merging of partitions but it starts increasing rapidly after the creation of 4 partitions in both network topologies. Therefore, the fog layer has been divided into 4 partitions for optimal network performance in both network topologies. In each partition, the fog node closest to the cloud server has been designated as the partition head and would be the first node to receive *Content* messages coming from servers. After partitioning, the performance of the DPPCOP has been analyzed for different threshold values (T_{FN}) in the Abilene-based network topology with $\alpha = 0.8$ and $\alpha = 1.1$. Fig. 4 shows the hit-ratio of the proposed DPPCOP scheme for threshold values varying between 1 to 7 and $\alpha = 0.8$. The optimal hit-ratio has been achieved with $T_{FN} = 2$. The hit ratio is also computed for $\alpha = 1.1$ on the Abilene-based network for different values of T_{FN} and the highest cache hit ratio is achieved when $T_{FN} = 1$ (hit-ratio $\approx$ 26%). Similarly, in US26-based network topology, the hit ratio is analyzed for different threshold values with $\alpha = 0.8$ and 1.1. In this network topology, the highest hit ratio is obtained when $T_{FN} = 2$ (hit-ratio $\approx$ 11%) and $T_{FN} = 1$ (hit-ratio $\approx$ 31%) for $\alpha = 0.8$ and 1.1 respectively. Although the computation of δ and T_{FN} is comparatively arbitrary and may vary for other fog networks, it offers a good foundation point to measure the performance of large-scale CCN-based fog networks.

To verify that the proposed DPPCOP scheme divided the network into optimal number of partitions, the hit ratio value is examined for the different number of network partitions with the above-discussed simulation parameters. The simulation results obtained on the Abilene-based network topology are plotted in Fig. 5 for $\alpha = 0.8$ with $T_{FN} = 2$, which shows that the highest network hit ratio is obtained when the network is divided into 4 partitions. Similarly, when $\alpha = 1.1$, the highest hit ratio on Abilene-based network is achieved, when network is divided into 4 partitions. Analogous results has been obtained for US26-based network topology. Thus, the obtained results also confirm that the proposed network partitioning scheme divides the Abilene and US26-based topologies into the optimal number of partitions.

5.2. Cache hit ratio

The cache hit ratio is one of the prime indicators to measure the QoS delivered by the caching schemes. When popular content is cached nearer to the requesters, then the majority of requests are fulfilled from the cache of fog nodes, which results in decreased cache miss rate. Therefore, the hit ratio improves with an increase in the accuracy of the content placement decisions. The in-network cache hit ratio has been computed using Eq. (5) [66], where n represents the number of fog nodes that have the content caching capability in the network.

$$Hit_Ratio = \frac{\sum_{i=1}^{n} Hit(FN_i)}{\sum_{i=1}^{n} Hit(FN_i) + \sum_{i=1}^{n} Miss(FN_i)} \tag{5}$$

Fig. 6 shows the average cache hit-ratio of the schemes when content popularity distribution follows the Zipf distribution with $\alpha = 0.8$. Initially, the hit ratio of all caching schemes is low in both network topologies because

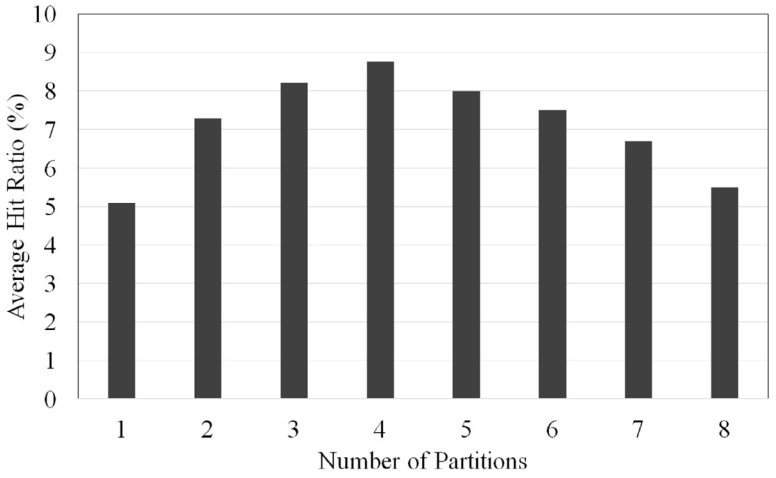
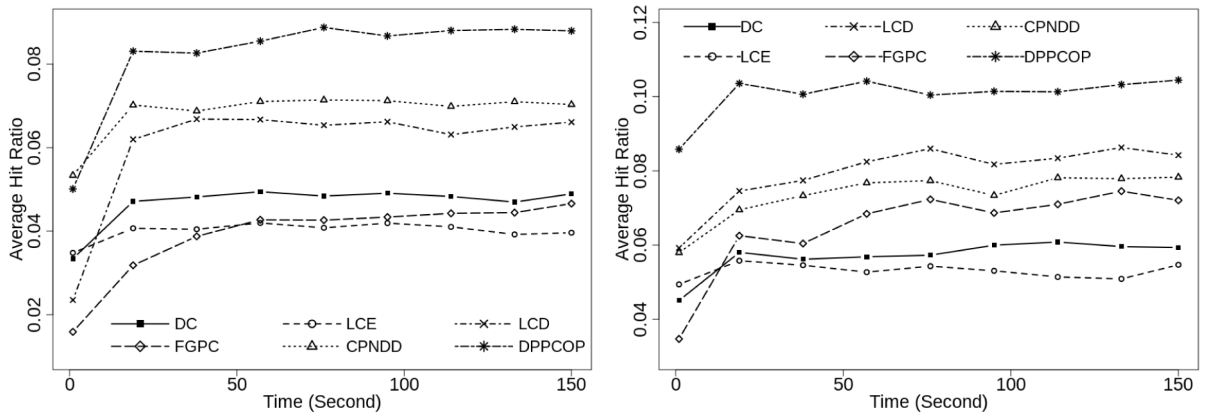


Fig. 5. Hit ratio observed for the proposed DPPCOP scheme with different number of network partitions and $(T_{FN}) = 2$.



(a) Average network hit ratio on Abilene-based topology with $\alpha = 0.8$

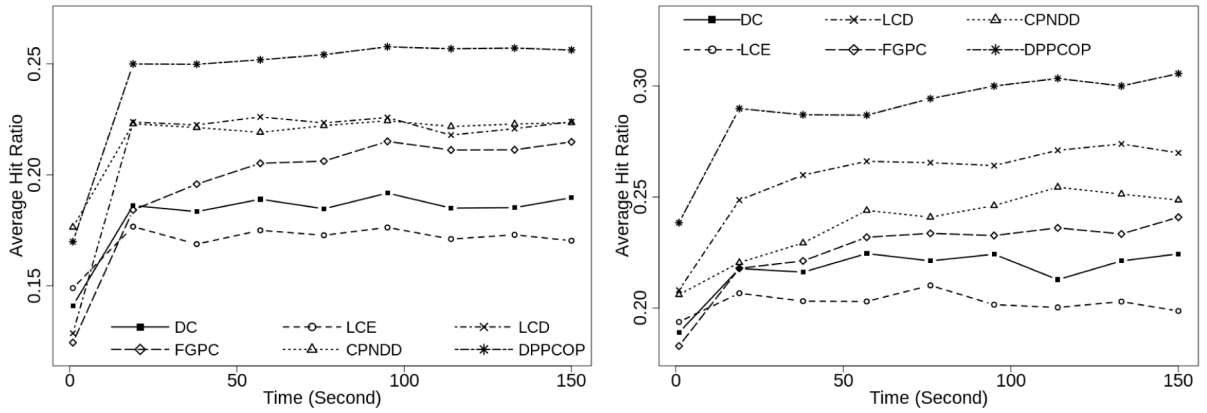
(b) Average network hit ratio on US26-based topology with $\alpha = 0.8$

Fig. 6. In-network cache hit-ratio of caching schemes on Abilene and US26-based topology with $|D| = 5000$, $|CS(FN_i)| = 50$, $\alpha = 0.8$ and $\lambda = 50$/s.

the cache of fog nodes is empty in the beginning. With content caching operations, the in-network hit-ratio rises sharply and begins saturated from 25th seconds onwards in both network topologies. During simulations, each end-user IoT device is generating 50 *Interest* messages per second ($\lambda = 50$) and thus, in the Abilene-based topology, nearly $50 \times 133 \times 25$ ($\lambda \times Number\ of\ IoT - devices \times simulation\ time$)($= 166250$) *Interest* messages are generated and processed in every 25 s using the caching scheme. Similarly, a large number of Interest messages are generated and processed in the US26-based network topology in each second. This also brings an interesting insight that the value of hit ratio saturates after processing of a "sufficient" number of *Interest* messages using the applied caching strategy. This happens because the caching schemes have identified and cached majority of the suitable contents at appropriate fog-nodes using their underlying heuristics in the initial duration of execution. As illustrated in Fig. 6(a) and Fig. 6(b), the saturation time for hit ratio is dependent on the number of *Interest* messages processed in the network and do not show much correlation with the underlying caching scheme.

In both network topologies, the DPPCOP scheme shows an improved cache hit ratio as compared to peer caching schemes. The cache hit ratio of all schemes does not increase uniformly with time due to the replacement of popular contents with unpopular ones. This causes a decrease in the hit ratio for some duration until the unpopular contents are again replaced by the frequently requested contents. The increased cache hit ratio also indicates that the content redundancy is minimized within the intra-partition fog nodes and the popular contents are efficiently distributed in the fog network.

In Fig. 7, it has been observed that the average hit-ratio of all caching strategies rises with an increase in the exponent value of Zipf distribution. When a smaller subset of contents are accessed more frequently ($\alpha = 1.1$), then a smaller



(a) Average network hit ratio on Abilene-based topology with $\alpha = 1.1$

(b) Average network hit ratio on US26-based topology with $\alpha = 1.1$

Fig. 7. In-network cache hit-ratio of caching schemes on Abilene and US26-based topology with $|D| = 5000$, $|CS(FN_i)| = 50$, $\alpha = 1.1$ and $\lambda = 50/s$.

cache size would be adequate to satisfy more *Interest* messages as compared to the scenario when the value of α is 0.8. As shown in Fig. 7(a) for Abilene-based topology, the hit-ratio of the DPPCOP scheme has been increased from 8.8% (shown in Fig. 6(a)) to 25.7% with an increase of α from 0.8 to 1.1. Fig. 7(a) shows that the DPPCOP caching scheme outperforms the peer strategies by showing 6.6%, 3.3%, 8.6%, 4.2% and 3.3% gain in hit ratio as compared to DC-based, LCD, LCE, FGPC, and CPNDD schemes on Abilene network. A similar improvement in hit ratio is also obtained on US26-based topology as shown in Fig. 7(b), where DPPCOP achieves 3.6% − 10.7% gain from the existing caching schemes.

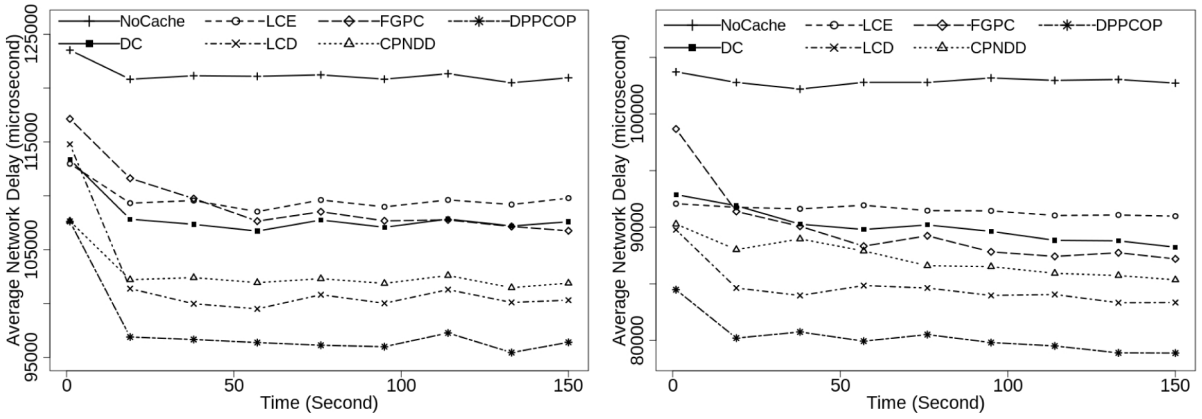
5.3. Network delay (in microseconds)

The network delay is computed as the time interval between the generation of the *Interest* message by the end-user and receiving of the requested content. During simulation executions, the delay metric is computed as the sum of transmission delay, processing delay, propagation delay, and queuing delay. It also includes the request retransmission latency, if the requested content is not received before the timeout of PIT entry. The average network delay has been determined as the average of this observed interval for all end-users devices. Figs. 8 and 9 shows the average network delay in the Abilene and US26-based topologies for $\alpha = 0.8$ and 1.1 respectively. As the DPPCOP uses content popularity and distance metrics along with network partitioning for caching decisions, the caching probability increases for popular contents with an increase in the distance from the content provider. Hence, the contents are placed near the end-users. As shown in Fig. 8(a), the DPPCOP shows a 3.9% to 20.8% reduction in network delay from peer caching strategies on the Abilene-based network, and analogous results are achieved for US26-based network as shown in Fig. 8(b). A decrease in the network delay metric for the proposed caching scheme also indicates that the content distribution is improved throughout the network. Thus, the cache space is comprehensively utilized and the popular contents are placed near the edges of the network. Due to this, the content requests are served with lesser delay as compared to the peer caching strategies.

Furthermore, when popularity distribution rises for a smaller subset of contents due to an increase in $\alpha(= 1.1)$, the average network delay decreases for the caching schemes as shown in Fig. 9 for both network topologies. In the Abilene-based network, the DPPCOP scheme demonstrates a 17.1% and 7.8% reduction in average network delay from the traditional LCE caching scheme and CPNDD strategy. Similarly, in the US26-based topology, the proposed DPPCOP scheme outperforms the existing caching schemes by showing a 6.4% to 19.1% reduction in average delay which is nearly 45% reduction from the *NoCache* strategy. The value of transmission, propagation, and queuing latencies is constant between any two network nodes for all the caching schemes during simulations. Hence, the reduction in average network delay also verifies that the proposed caching scheme does not increase the processing overhead in the fog nodes during *Content* message forwarding and reduces the average network delay through efficient content placement decisions.

5.4. Network hop-count

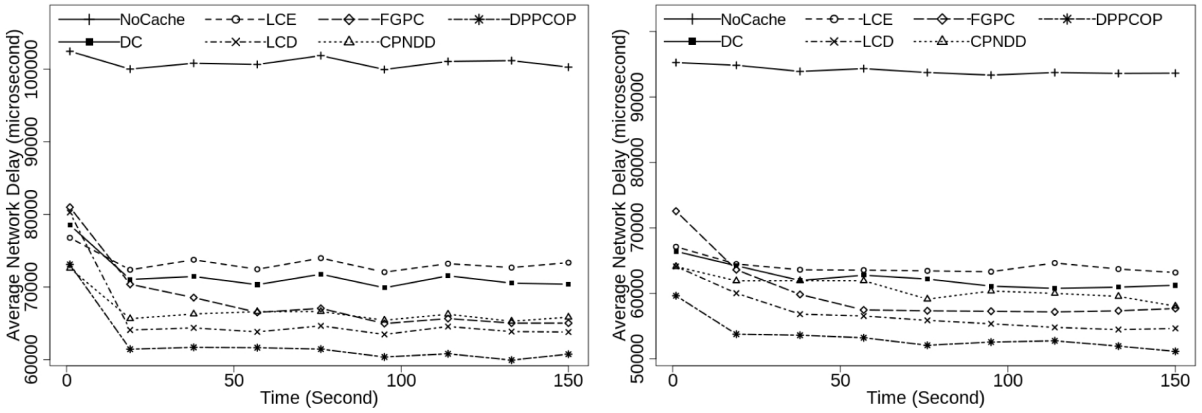
The average network hop-count is determined as the sum of the number of hops that are traversed by the *Interest* message and its matching *Content* message from the end-user to the content provider and vice versa [32]. Generally, the hop-count metric is proportional to the network delay. However, the value of hop-count metric is static as compared to the network delay parameter, which is dynamic and depends on several real-time factors of the network such as physical distance, varying queuing delay, processing delay and traffic conditions in the path. Thus, the hop count is a



(a) Average network delay (in microseconds) on Abilene-based topology with $\alpha = 0.8$

(b) Average network delay (in microseconds) on US26-based topology with $\alpha = 0.8$

Fig. 8. Average network delay (in microseconds) of caching schemes on Abilene and US26-based topology with $|D| = 5000$, $|CS(FN_i)| = 50$, $\alpha = 0.8$ and $\lambda = 50/s$.



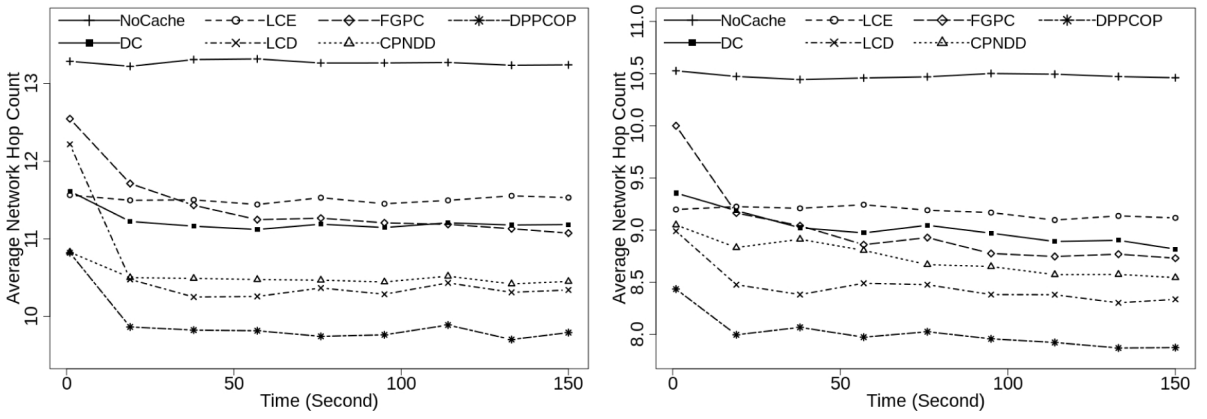
(a) Average network delay (in microseconds) on Abilene-based topology with $\alpha = 1.1$

(b) Average network delay (in microseconds) on US26-based topology with $\alpha = 1.1$

Fig. 9. Average network delay (in microseconds) of caching schemes on Abilene and US26-based topology with $|D| = 5000$, $|CS(FN_i)| = 50$, $\alpha = 1.1$ and $\lambda = 50/s$.

stable measurement metric which is not influenced by the dynamic factors of network. The value of hop-count metric decreases when the content is placed nearer to the end-users. Fig. 10 shows the average network hop-count observed by the end-user devices when $\alpha = 0.8$. The simulation results obtained on the Abilene-based network illustrate that the DPPCOP outperforms the existing scheme by demonstrating a 5.3% to 24.8% reduction in average network hop-count from LCE, LCD, DC-based, FGPC, and CPNDD strategies. As the network size is smaller in the US26-based topology from the Abilene-based network, the value of average hop count is also less in this network as shown in Fig. 10(b). In this scenario, the DPPCOP strategy reduces the average hop count up to 5.6% and 24.7% from the LCD caching approaches.

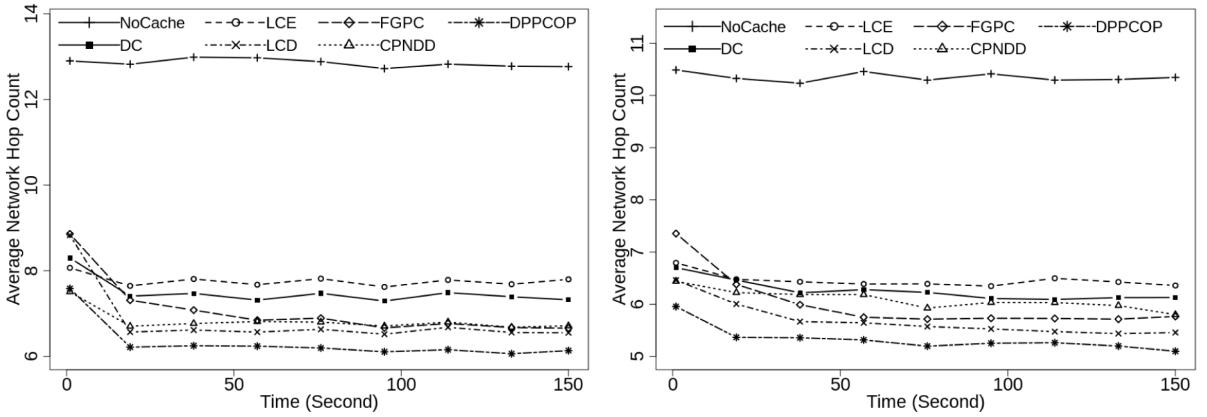
Analogous to previous results, when the exponent value in the Zipf distribution becomes 1.1, the DPPCOP scheme reduces the average network hop-count up to 51.2% from peer strategies in the Abilene-based network as shown in Fig. 11(a). Fig. 11(b) illustrates the average network hop count for retrieving the requested contents in US26-based network which shows the superiority of DPPCOP approach on peer competing strategies. In contrast to the network delay metric (discussed in Section 5.3), the reduction in hop-count metric concludes that the contents are actually accessed from the nearby fog nodes using the proposed DPPCOP caching scheme because this metric does not depend on the varying computational delay, queuing and dynamic traffic conditions in the network.



(a) Average network hop-count on Abilene-based topology with $\alpha = 0.8$

(b) Average network hop-count on US26-based topology with $\alpha = 0.8$

Fig. 10. Average network hop-count of caching schemes to access contents on Abilene and US26-based topology with $|D| = 5000$, $|CS(FN_i)| = 50$, $\alpha = 0.8$ and $\lambda = 50/s$.



(a) Average network hop-count on Abilene-based topology with $\alpha = 1.1$

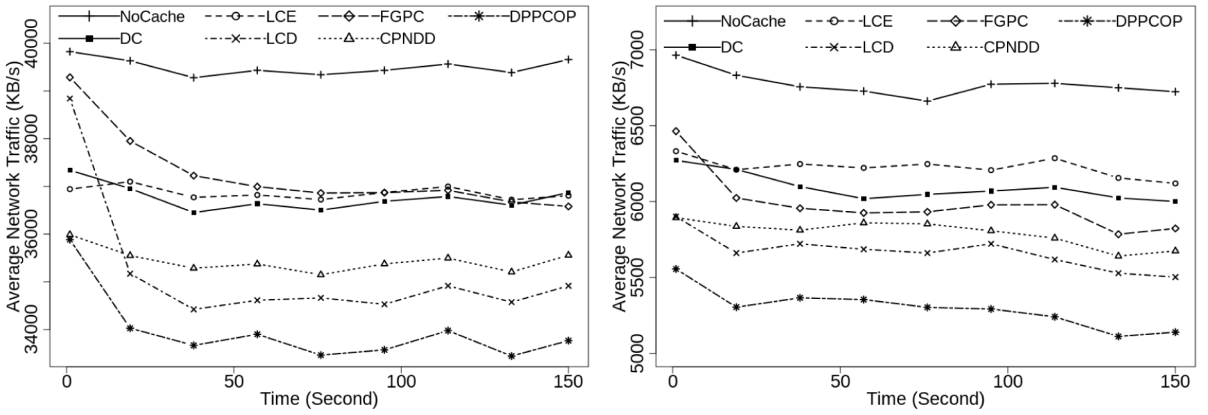
(b) Average network hop-count on US26-based topology with $\alpha = 1.1$

Fig. 11. Average network hop-count of caching schemes to access contents on Abilene and US26-based topology with $|D| = 5000$, $|CS(FN_i)| = 50$, $\alpha = 1.1$ and $\lambda = 50/s$.

5.5. Network traffic load

The value of average network traffic load metric has been determined as the sum of network traffic (in KB) observed on the network links in per unit time (per second). It indicates the utilization of available network bandwidth. Therefore, the network traffic load tends to decrease, with an improvement in the accuracy of content placement decisions. Due to the underlying heuristics in the DPPCOP scheme, the popular contents are placed near the end-users. Hence, when $\alpha = 0.8$, the DPPCOP scheme decreases the average load of network traffic by 8.3% and 8.5% from LCE and DC-based caching schemes in the Abilene-based network as shown in Fig. 12(a). The FGPC scheme that considers the content popularity during content placement decisions, also shows significant performance gain with an increase in the exponent value of Zipf distribution as shown in Fig. 13. When α is increased to 1.1, the proposed DPPCOP scheme outperforms the existing peer strategies in both the network topologies in terms of average traffic load on the network links. As demonstrated in Figs. 13(a) and 13(b), the reduction of average network traffic load varies between 4% − 26.8% and 7.7% − 42.95% from peer schemes in the Abilene and US26-based networks respectively.

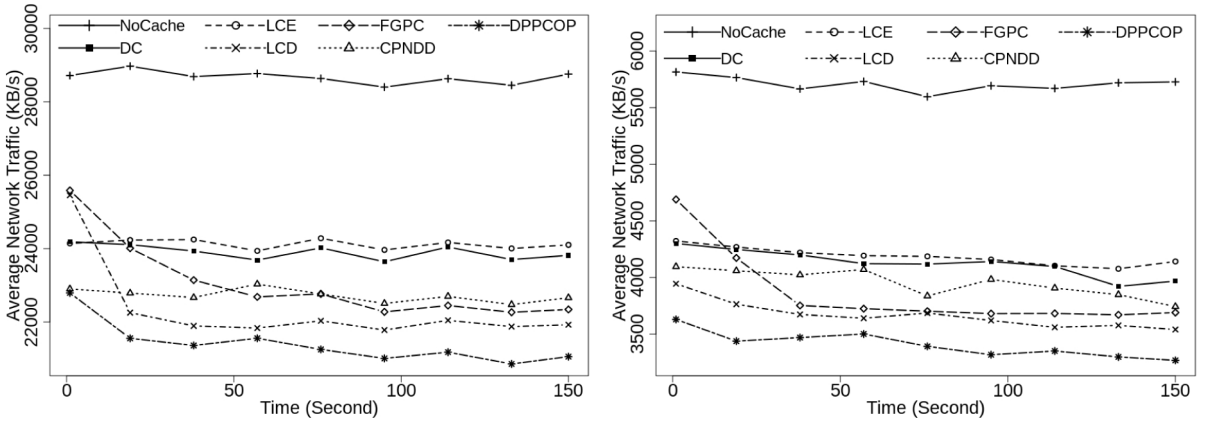
Based on the obtained simulation results on Abilene and US26-based network topologies, it has been observed that the proposed DPPCOP scheme significantly improves the cache hit probability near the end-user IoT devices. Using the proposed approach, the required contents are accessed with lesser delay and the network bandwidth consumption is also



(a) Average network traffic load (in KB/second) on Abilene-based topology with $\alpha = 0.8$

(b) Average network traffic load (in KB/second) on US26-based topology with $\alpha = 0.8$

Fig. 12. Average network traffic load (in KB/s) of caching schemes on Abilene and US26-based topology with $|D| = 5000$, $|CS(FN_i)| = 50$, $\alpha = 0.8$ and $\lambda = 50/s$.



(a) Average network traffic load (in KB/second) on Abilene-based topology with $\alpha = 1.1$

(b) Average network traffic load (in KB/second) on US26-based topology with $\alpha = 1.1$

Fig. 13. Average network traffic load (in KB/s) of caching schemes on Abilene and US26-based topology with $|D| = 5000$, $|CS(FN_i)| = 50$, $\alpha = 1.1$ and $\lambda = 50/s$.

reduced. Therefore, the DPPCOP scheme is suitable for the IoT applications such as gaming, health emergencies, and public safety, which need faster content access with lesser network traffic load.

6. Conclusion

In CCN-based fog networks, the content caching capability in the fog nodes raises new challenges to efficiently utilize the available caching resources and obtain improved network performance. In this paper, a novel content caching scheme, called DPPCOP has been proposed that dynamically partitioned the fog networks into the optimal number of partitions. The intra-partition fog nodes takes collaborative caching decisions to avoid excessive content redundancy and improves content distributions in the network. The partitioning scheme does not incur any significant communication or computation overhead in the network. During content caching decisions, a large size popularity window is used to find popular content requests, which is jointly considered with the distance of the content provider during caching decisions. The DPPCOP improves the QoS during content retrieval as the proposed heuristics increase the caching probability of popular contents near the end-user devices. Exhaustive simulations are performed for a wide variety of content request patterns that follows Zipf distribution with $\alpha = 0.8$ and $\alpha = 1.1$. The results demonstrate that the DPPCOP increases the cache hit ratio and reduces the delay, hop-count, and network traffic load parameters from the existing caching strategies

on the Abilene and US26-based network topologies. Hence, the DPPCOP scheme is suitable for modern IoT applications such as gaming and health emergencies that require content retrieval with low latency and efficient utilization of the available network bandwidth.

In future, the influence of end-users mobility can be examined on the performance of the DPPCOP scheme. Additional parameters such as node centrality can also be considered during caching operations to enhance the caching performance for next-generation network architectures.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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